# Exercise Sheet 10

The conventions are those of Lecture 10: $a_{ij} = \alpha_j(h_i)$, the forms $J_\pm$ and $S$, and an element $\operatorname{diag}(t, -t)$ or $\operatorname{diag}(0, t, -t)$ of $\mathfrak{h}$ named by its vector $t = \sum t_i e_i$. Exercises 10.1 and 10.2 build classical root tables, and Exercise 10.3 computes determinants of Cartan matrices. Exercise 10.4 treats selected exclusions. Exercises 10.5–10.7 concern Serre relations and low rank.

**Exercise 10.1 (A root table for $B_3$).** Use the model $\mathfrak{so}_7$ of the lecture with $\alpha_1 = \varepsilon_1 - \varepsilon_2$, $\alpha_2 = \varepsilon_2 - \varepsilon_3$ and $\alpha_3 = \varepsilon_3$.

(i) Write the nine positive roots in terms of the simple roots.

(ii) Compute the Cartan matrix and describe the Dynkin diagram.

(iii) Check $|R|$ and $\dim \mathfrak{so}_7$. Which positive root has the largest coefficient sum?

**Exercise 10.2 ($C_3$, transposes and a root count).**

(i) Compute $A(C_3)$ with $\alpha_3 = 2\varepsilon_3$ and compare it with $A(B_3)$.

(ii) Humphreys puts $2(\alpha_i, \alpha_j)/(\alpha_j, \alpha_j)$ in position $(i, j)$. Which of our matrices appears in his table for $B_3$?

(iii) In the $C_n$ model, show that the coroot of $\varepsilon_i + \varepsilon_j$ $(i < j)$ has vector $e_i + e_j$, and that of $2\varepsilon_i$ has vector $e_i$.

(iv) Write $\langle \beta, \alpha^\vee \rangle = \beta(h_\alpha)$. In $B_3$ and in $C_3$, count the roots $\alpha$ for which some root $\beta \neq \pm\alpha$ has $|\langle \beta, \alpha^\vee \rangle| = 2$. Deduce that $R(B_3) \not\cong R(C_3)$.

**Exercise 10.3 (Determinants).** Let $T_k$ be a $k \times k$ tridiagonal matrix with diagonal entries 2, entries $b_1, \dots, b_{k-1}$ above the diagonal and $c_1, \dots, c_{k-1}$ below it. Put $\det T_0 = 1$.

(i) For $k \geq 2$, prove $\det T_k = 2 \det T_{k-1} - b_{k-1}c_{k-1} \det T_{k-2}$.

(ii) Deduce $\det A(A_\ell) = \ell + 1$ and $\det A(B_\ell) = \det A(C_\ell) = 2$ for $\ell \geq 2$.

(iii) Compute $\det A(D_4)$.

(iv) Explain why the determinant of every Cartan matrix is positive.

**Exercise 10.4 (Selected exclusions).** Use the form $q_\Gamma$ of Lecture 10. Show that no Dynkin diagram contains any of the following graphs after deleting nodes.

(i) A chain of seven nodes with an eighth node joined to the middle one, all edges single.

(ii) Nodes $a, b, c, d, e$ with single edges $a$–$c$, $b$–$c$, $c$–$d$ and a double edge $d$–$e$.

(iii) A chain of five nodes $1, \dots, 5$ with single edges, except a double edge between nodes 3 and 4.

(iv) Why is the direction of the arrows irrelevant here?

To find the vectors, use the search method of Lecture 10, starting at a node that meets only one edge.

**Exercise 10.5 (Serre relations in $\mathfrak{sp}_4$).** In the lecture's model with $n = 2$, take the root vectors $e_1 = X_{12} = E_{12} - E_{43}$ and $e_2 = E_{24}$, with coroots $h_1, h_2$.

(i) Write $h_1, h_2$ as diagonal matrices. Verify $[h_1, e_2] = a_{12}e_2$ and $[h_2, e_1] = a_{21}e_1$.

(ii) Compute $(\operatorname{ad} e_1)^k(e_2)$ for $k = 1, 2, 3$ and $(\operatorname{ad} e_2)^k(e_1)$ for $k = 1, 2$. Compare with the exponents $1 - a_{ij}$.

**Exercise 10.6 ($D_3$ and $A_3$ explicitly).** For $\mathfrak{sl}_4$ let $f_1, \dots, f_4$ be the diagonal coordinate functionals and $\beta_a = f_a - f_{a+1}$. For $D_3$ use $\alpha_1 = \varepsilon_1 - \varepsilon_2$, $\alpha_2 = \varepsilon_2 - \varepsilon_3$, $\alpha_3 = \varepsilon_2 + \varepsilon_3$. These $f_a$ are not the generators $f_i$ of the Serre relations, and $f_1 + f_2 + f_3 + f_4 = 0$ on the traceless diagonal matrices. Define a linear map $\varphi$ from the dual of the Cartan subalgebra of $\mathfrak{so}_6$ to that of $\mathfrak{sl}_4$ by $\varphi(\alpha_2) = \beta_1$, $\varphi(\alpha_1) = \beta_2$, $\varphi(\alpha_3) = \beta_3$.

(i) Compute $\varphi(\varepsilon_1)$, $\varphi(\varepsilon_2)$, $\varphi(\varepsilon_3)$.

(ii) Show that $\varphi$ maps the 12 roots of $D_3$ onto the 12 roots of $A_3$.

(iii) Explain why $\varphi$ is an isomorphism of root systems.

**Exercise 10.7 (Rank two).**

(i) Show that every $2 \times 2$ Cartan matrix is, after renumbering, one of four matrices, and name the types.

(ii) Find the angle between the simple roots and their length ratio in each case.

(iii) For the $G_2$ base of the lecture, show from the list of roots that $\alpha_2 + k\alpha_1 \in R$ exactly for $0 \le k \le 3$, and that $\alpha_1 + k\alpha_2 \in R$ exactly for $k \in \{0, 1\}$. Compare with the Serre exponents.